cp_9a

1

Print the Digits of a Number | 30/30

Easy

Write a program that reads a number and prints all the digits of the given number separated by a space.

Input

The input will be a single line containing a positive integer.

Output

The output should be a single line containing a string that has all the digits of the given number separated by a space.

Explanation

For example, if the given number is **165**,

- The digits of the number **165** are **1**, **6**, and **5**.
- Print all the digits of the number separated by a space.

The output should be **1 6 5**.

Sample Input 1

```
165
```

Sample Output 1

```
1 6 5
```

Sample Input 2

```
45866
```

Sample Output 2

```
4 5 8 6 6
```

code

```
n=int(input())
n_str=str(n)
final=""
for i in range(len(n_str)):
    val=n_str[i]+" "
    final=final+val
print(final)
```

2

Sum of Digits 35/35

Write a program to print the sum of all the digits in the given number.

Input

The input will be a single line containing a positive integer.

Output

The output should be a single line containing the sum of the digits in the given number.

Explanation

For example, if the given number is 151893, your code should print the sum of all the digits in the given number (1 + 5 + 1 + 8 + 9 + 3), which is 27.

For example, if the given number is 692, your code should print the sum of all the digits in the given number (6 + 9 + 2), which is 17.

Sample Input 1

```
151893
```

Sample Output 1

```
27
```

Sample Input 2

```
692
```

Sample Output 2

```
17
```

code

```
n=int(input())
n_str=str(n)
n_len=len(n_str)
sum=0
for i in range(n_len):
    sum=sum+int(n_str[i])
print(sum)
```

3

Factorial | 30/30

Write a program to print the factorial of `N` .Factorial is the product of all positive integers less than or equal to `N` .

Note

The **factorial** of **0** is **1**.

Input

The input will be a single line containing a positive integer representing `N` .

Output

The output should be a single line containing the factorial of the given number `N` .

Explanation

For example, if the given number is `N = 4` ,

- The factorial of **4** is **24** . `(4 × 3 × 2 × 1 = 24)`

So, the output should be **24**.

For example, if the given number is `N = 7` ,

- The factorial of **7** is **5040**. `(7 × 6 × 5 × 4 × 3 × 2 × 1 = 5040)`

So the output should be **5040**.

Sample Input 1

```
4
```

Sample Output 1

```
24
```

Sample Input 2

```
7
```

Sample Output 2

```
5040
```

code

```
n=int(input())
product=1
```

```
for i in range(1,n+1):
    product=product*n
    n=n-1
print(product)
```

```
4
```

Sum of Cubes | 30/30

Write a program that reads a number `N` , and prints the sum of the cubes of numbers from **1** to `N` .

Note

The cube of a number `X` is `X` power **3**.(X^3)

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing an integer that is the sum of the cubes of numbers from **1** to `N` .

Explanation

For example, if the given number is `N = 3` ,

- Numbers from **1** to **3** are **1**, **2**, and **3**.
- The cubes of numbers from **1** to **3** are **1**, **8**, and **27**.
- The sum of the cube of numbers from **1** to **3** is **36**. (**1 + 8 + 27 = 36**)

Number (X)	Cube of Number (X^3)	Previous Sum	Previous Sum + (X^3) = Current Sum
1	1	0	0 + 1 = 1
2	8	1	1 + 8 = 9
3	27	9	9 + 27 = 36

The output should be **36**.

Sample Input 1

```
3
```

Sample Output 1

```
36
```

code

```
n=int(input())
sum_of_cubes=0
for i in range(1,n+1):
    cube=i**3
```

```
    sum_of_cubes=sum_of_cubes+cube
print(sum_of_cubes)
```

5

Sum and Average of the 10 Inputs | 30/30

Write a program that reads 10 inputs and prints the sum and average of the given 10 inputs.

Note

The average of N numbers can be calculated as,
Average = Sum of N numbers / N

Example: If 3 numbers are 2, 3, and 7.

The average of 3 numbers is,
Average = (2 + 3 + 7) / 3 = 4.0

Input

The 10 lines of input contain integers.

Output

The first line of output should be an integer that is the sum of the given 10 inputs.
The second line of output should be a float that is the average of the given 10 inputs.

Explanation

For example, if the given 10 inputs are 4, 7, 14, 25, 1, 8, 24, 38, 99, and 10,

- The sum of 4, 7, 14, 25, 1, 8, 24, 38, 99, and 10 is 230. (`4 + 7 + 14 + 25 + 1 + 8 + 24 + 38 + 99 + 10 = 230`)
- The average of 4, 7, 14, 25, 1, 8, 24, 38, 99, and 10 is 23.0. (`230 / 10 = 23.0`)

code

```
sum=0
for i in range(10):
    n=int(input())
```

```
    sum=sum+n
avg=sum/10
print(sum)
print(avg)
```

```
6
```

Sum of Squares from M to N | 30/30

Easy

Write a program that reads two numbers `M` and `N`, and prints the sum of squares of numbers from `M` to `N`.

Input

The first line of input contains an integer representing `M`.
The second line of input contains an integer representing `N`.

Output

The output should be a single line containing an integer that is the sum of squares of numbers from `M` to `N`.

Explanation

For example, if the given numbers are `M = 2` and `N = 4`,

- Numbers from **2** to **4** are **2**, **3**, and **4**.
- The squares of numbers from **2** to **4** are **4**, **9**, and **16**.
- The sum of the squares of numbers from **2** to **4** is **29**. (**4 + 9 + 16 = 29**)

Number (X)	Square of Number (X^2)	Previous Sum	Previous Sum + (X^2) = Current Sum
2	4	0	0 + 4 = 4
3	9	4	4 + 9 = 13
4	16	13	13 + 16 = 29

The output should be **29**.

Sample Input 1

```
2
4
```

Sample Output 1

```
29
```

code

```
m=int(input())
n=int(input())
sum=0
for i in range(m,n+1):
```

```
    squares=i**2
    sum=sum+squares
print(sum)
```

7

Print Even Numbers up to N | 35/35

Easy

Write a program that reads a number `N` and prints even numbers from **1** to `N` .

Input

The input will be a single line containing an integer representing `N` .

Output

The output should contain all the even numbers from **1** to `N` , each on a new line.

Explanation

For example, if the given number is `N = 7` ,

- The numbers from **1** to **7** are **1**, **2**, **3**, **4**, **5**, **6**, and **7**.
- The even numbers among numbers from **1** to **7** are **2**, **4**, and **6**.

The output should be,

```
2
4
6
```

Sample Input 1

```
7
```

Sample Output 1

```
2
4
6
```

code

```python
n=int(input())
for i in range(1,n+1):
    evens=(i%2)
    if evens==0:
        print(i)
```

8

Print Odd Numbers from M to N | 40/40

Easy

Write a program that reads two numbers `M` and `N` and prints odd numbers from `M` to `N` separated by a space.

Input

The first line of input contains an integer representing `M` .
The second line of input contains an integer representing `N` .

Output

The output should be a single line containing a string that has odd numbers from `M` to `N` separated by a space.

Explanation

For example, if the given numbers are `M = 5` and `N = 10` ,

- Numbers from **5** to **10** are **5**, **6**, **7**, **8**, **9**, and **10**.
- The odd numbers among numbers from **5** to **10** are **5**, **7**, and **9**.

The output should be **5 7 9**.

Sample Input 1

```
5
10
```

Sample Output 1

```
5 7 9
```

Sample Input 2

```
9
21
```

Sample Output 2

```
9 11 13 15 17 19 21
```

code

```
m=int(input())
n=int(input())
odds=""
for i in range(m,n+1):
    odd=i%2
    if odd==1:
        odds=odds+str(i)+" "
print(odds)
```

9

Sum of Even Numbers Upto N | 30/30

Write a program that reads a number `N` and prints the sum of the even numbers from **1** to `N` .

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing an integer that is the sum of the even numbers from **1** to `N` .

Explanation

For example, if the given number is `N = 6` ,

- The numbers from **1** to **6** are **1**, **2**, **3**, **4**, **5**, and **6**.
- The even numbers among the numbers from **1** to **6** are **2**, **4**, and **6**.
- The sum of even numbers from **1** to **6** is **12**. (`2 + 4 + 6 = 12`)

Number	Previous Sum	Previous Sum + Number = Current Sum
2	0	0 + 2 = 2
4	2	2 + 4 = 6
6	6	6 + 6 = 12

The output should be **12**.

Sample Input 1

```
6
```

Sample Output 1

```
12
```

code

```
n=int(input())
sum=0
for i in range(1,n+1):
    even=i%2
    if even==0:
        sum=sum+i
```

```
print(sum)
```

10

Sum of Odd Numbers Upto N | 30/30

Write a program that reads a number `N` and prints the sum of the odd numbers from **1** to `N` .

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing an integer that is the sum of the odd numbers from **1** to `N` .

Explanation

For example, if the given number is `N = 6` ,

- The numbers from **1** to **6** are **1**, **2**, **3**, **4**, **5**, and **6**.
- The odd numbers among numbers from **1** to **6** are **1**, **3**, and **5**.
- The sum of odd numbers from **1** to **6** is **9**. (`1 + 3 + 5 = 9`)

Number	Previous Sum	Previous Sum + Number = Current Sum
1	0	0 + 1 = 1
3	1	1 + 3 = 4
5	4	4 + 5 = 9

The output should be **9**.

code

```
n=int(input())
sum=0
for i in range(n+1):
    rem=i%2
```

```
    if rem==1:
        sum=sum+i
print(sum)
```

11

Sum of Even Numbers from M to N | 30/30

Write a program that reads two numbers `M` and `N` and prints the sum of the even numbers from `M` to `N`.

Input

The first line of input contains an integer representing `M`.
The second line of input contains an integer representing `N`.

Output

The output should be a single line containing an integer that is the sum of the even numbers from `M` to `N`.

Explanation

For example, if the given two numbers are `M = 2` and `N = 6`,

- The numbers from **2** to **6** are **2**, **3**, **4**, **5**, and **6**.
- The even numbers among the numbers from **2** to **6** are **2**, **4**, and **6**.
- The sum of even numbers from **2** to **6** is **12**. (`2 + 4 + 6 = 12`)

Number	Previous Sum	Previous Sum + Number = Current Sum
2	0	0 + 2 = 2
4	2	2 + 4 = 6
6	6	6 + 6 = 12

The output should be **12**.

code

```
m=int(input())
n=int(input())
sum=0
for i in range(m,n+1):
```

```
    rem=i%2
    if rem==0:
        sum=sum+i
print(sum)
```

12

Product of Odd Numbers from M to N | 30/30

Easy

Write a program that reads two numbers `M` and `N`, and prints the product of odd numbers from `M` to `N`.

Input

The first line of input contains an integer representing `M`.
The second line of input contains an integer representing `N`.

Output

The output should be a single line containing an integer that is the product of the odd numbers from `M` to `N`.

Explanation

For example, if the given numbers are `M = 2` and `N = 7`,

- Numbers from **2** to **7** are **2**, **3**, **4**, **5**, **6**, and **7**.
- The odd numbers among numbers from **2** to **7** are **3**, **5**, and **7**.
- The product of odd numbers from **2** to **7** is **105**. (`3 * 5 * 7 = 105`)

Number	Previous Product	Previous Product * Number = Current Product
3	1	1 * 3 = 3
5	3	3 * 5 = 15
7	15	15 * 7 = 105

The output should be **105**.

code

```
m=int(input())
n=int(input())
```

```
prod=1
for i in range(m,n+1):
    rem=i%2
    if rem==1:
        prod=prod*i
print(prod)
```

13

Sum of Odd Numbers from M to N | 30/30

Given two integers M and N, write a program to print the sum of the odd numbers from M to N.

Input

The first line of input will contain an integer.
The second line of input will contain an integer.

Output

The output should be a single line containing the sum of the odd numbers from M to N.

Explanation

For example, if the given numbers are 5 and 13, the output should be the sum of the odd numbers from 5 to 13 (5 + 7 + 9 + 11 + 13), which is 45.

Sample Input 1

```
5
13
```

Sample Output 1

```
45
```

Sample Input 2

```
10
20
```

Sample Output 2

```
75
```

code

```
m=int(input())
n=int(input())
sum=0
```

```
for i in range(m,n+1):
    rem=i%2
    if rem==1:
        sum=sum+i
print(sum)
```